

Notes: Philipson and Becker (JPE, 1998)
Old-Age Longevity and Mortality-Contingent Claims

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Objects of interest	4
3.2	Longevity-related behavior	4
3.3	Public old-age programs	5
3.4	Empirical illustration: Civil Service Retirement System	5
4	Models and empirical specifications	5
4.1	Baseline retirement problem without mortality-contingent income	5
4.2	Endogenous longevity	6
4.3	Mortality-contingent wealth	6
4.4	Private annuity pricing and moral hazard	7
4.5	Public annuities	7
4.6	Morbidity-contingent annuity allocation	7
4.7	PAYG public annuities and private savings	8
4.8	Public annuities and public health care	8
4.9	Empirical decomposition of annuity value	9
5	Identification and empirical design	9
5.1	What is identified conceptually	9
5.2	Private versus public pricing	10
5.3	Evidence and limitations in the CSRS illustration	10
5.4	Main empirical obstacle	10
6	Tables and figures	11
6.1	Figures 1 and 2: The mechanism and the survival evidence	11
6.2	Figure 3: Incentive effects by monthly annuity increase	12

7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13
8.2	Economic intuition	13
8.3	Incremental contribution revisited	13
8.4	Policy implications	13
8.5	Limitations	14
8.6	Takeaway	14

1 Big picture

- **Question.** How do mortality-contingent claims—especially annuities and life insurance—change savings, health investment, and longevity when mortality is not fixed but partly chosen through behavior?
- **Core idea.** An annuity pays more total income the longer a person lives. Once longevity can be affected by health spending, lifestyle, diet, care, or family time transfers, annuities create a piece-rate incentive for survival.
- **Main departure from standard analysis.** Conventional old-age support analysis often treats mortality as exogenous. Philipson and Becker instead make longevity endogenous, so Social Security and Medicare-like programs change not only wealth and consumption, but also the incentive to invest in living longer.
- **Private-market insight.** Private annuity markets face a moral hazard problem: individuals who invest in longevity impose higher expected payout costs on other annuitants unless premia can fully adjust for longevity-related behavior.
- **Public-program insight.** Public annuities are mandatory and typically not risk-rated by detailed longevity behavior. This makes the longevity incentive stronger than in private markets and can make longevity excessive relative to a lump-sum old-age transfer.
- **Health-program interaction.** Public annuities and public health insurance reinforce each other. Annuities raise the benefit of living longer, while health subsidies reduce the cost of living longer. Each program can therefore increase the fiscal cost of the other.
- **Savings implication.** With endogenous longevity, public old-age savings and private savings need not be substitutes. If public annuities induce longer retirement lives, private individuals may need to save more to finance the longer life.
- **Empirical illustration.** The paper uses Civil Service Retirement System data to show that survival is higher among men with larger monthly annuities and that a nontrivial part of the increase in annuity value may come from induced longevity.
- **Economic takeaway.** Programs such as Social Security do not merely transfer resources to old age. They also change the return to health investment, the duration of retirement, private savings needs, and the interaction between health care and income support.

2 Incremental contribution of the paper

- **From exogenous to endogenous mortality.** The paper’s central contribution is to bring endogenous longevity into the economics of annuities and old-age social insurance.
- **A new moral-hazard channel.** Standard annuity theory emphasizes insurance against outliving assets. This paper emphasizes that annuities also create an incentive to extend life, because the payout stream continues while alive.
- **Quality-versus-quantity trade-off.** The paper reframes annuities as changing the trade-off between living well and living long. A lump-sum retiree who lives longer spreads a fixed stock of wealth over more years; an annuitized retiree receives extra income when life is extended.

- **Public finance contribution.** The analysis shows why mandatory public annuities can have unrecognized welfare and distributional implications: they can induce excessive longevity investment and make benefits more regressive because higher annual benefits are received for longer by longer-lived individuals.
- **Interaction of Social Security and Medicare.** The paper highlights a positive feedback between old-age income support and health care subsidies. This is a distinctive public-finance implication of endogenous longevity.
- **Morbidity incompleteness.** The paper notes that standard annuities condition on mortality, not morbidity. If consumption is less valuable when sick, annuity income can be too high in bad-health states and may encourage excessive treatment rather than prevention.
- **Aggregate-savings implication.** The paper challenges the standard view that public pensions necessarily crowd out private savings. If annuities extend retirement life, private savings may increase.
- **Empirical measurement contribution.** The CSRS exercise does not claim a full causal estimate, but it shows how to decompose the value of a larger annuity into a quantity component and an incentive component due to induced longevity.

3 Data and measurement

3.1 Objects of interest

- The main theoretical object is a **mortality-contingent claim**: a financial arrangement whose total payoff depends on the length of life.
- The main examples are annuities, life insurance, and survivor benefits in pension plans.
- Annuities pay while alive, so the total payoff rises with longevity.
- Life insurance and survivor benefits can create different incentives because payments are tied to death or survival of another person.

Interpretation: The paper studies the incentive properties of claims whose payoff is mechanically tied to mortality. These claims do not only insure mortality risk; they also change behavior when longevity is partly choice-sensitive.

3.2 Longevity-related behavior

- Longevity is represented as a function of resources or behaviors, denoted conceptually by $T(M)$.
- M includes direct medical spending, home care, long-term care, diet, lifestyle choices, prescriptions, experimental therapies, and time transfers from family members.
- The paper treats $T(M)$ as increasing and concave: additional longevity investments raise life length but with diminishing returns.
- This formulation abstracts from detailed health production in order to focus on the economic incentives created by mortality-contingent payoffs.

Interpretation: The model is deliberately simple. Its purpose is to show how annuity incentives enter the first-order condition for longevity investment.

3.3 Public old-age programs

- The paper is motivated by the postwar expansion of mandatory old-age income programs and public health care programs.
- In the United States, the relevant institutional analogues are Social Security and Medicare.
- Public annuities differ from private annuities because participation is mandatory and premia are not risk-rated in the same detailed behavioral way as competitive insurance contracts.
- Public health insurance is described as “sick insurance” because it pays more in bad-health states, while annuity income continues regardless of morbidity state so long as the individual remains alive.

Interpretation: The public-sector setting is where the mechanism is most important, because pricing discipline is weak and the programs are very large.

3.4 Empirical illustration: Civil Service Retirement System

- The empirical section uses data reported by Virga on the U.S. Civil Service Retirement System (CSRS).
- CSRS covered federal employees hired before 1983.
- The data cover fiscal years 1988–1994 and report mortality hazard rates by salary and annuity cells.
- The paper focuses on males who have reached age 57 and constructs survival functions by monthly annuity level.
- The annuities are indexed and mandatory, so the authors argue that selection is likely less important than in a voluntary private annuity market, although they still treat the evidence as illustrative rather than causal.

Interpretation: The CSRS data are used to demonstrate how one can measure annuity-induced incentive components. The paper does not claim to fully solve identification of annuity effects on longevity.

4 Models and empirical specifications

4.1 Baseline retirement problem without mortality-contingent income

A retired individual has retirement wealth W , remaining lifetime T , and instantaneous utility $U(c)$. The indirect utility from optimally allocating consumption over life is:

$$V(W, T) = \max_{c(t)} \int_0^T U(c(t)) dt \quad \text{subject to} \quad \int_0^T c(t) dt \leq W.$$

With no discounting and concave utility, the individual smooths consumption equally across years:

$$c(t) = \frac{W}{T}, \quad V(W, T) = TU\left(\frac{W}{T}\right).$$

Interpretation: If wealth is a lump sum, living longer means spreading the same resources over more years. Longevity has a quantity benefit, but it lowers annual consumption.

4.2 Endogenous longevity

Let longevity depend on health or longevity investment M , so remaining life is $T(M)$. Without mortality-contingent wealth, the individual chooses M to maximize:

$$V(W - M, T(M)).$$

The basic trade-off is:

$$V_T T_M = V_W.$$

- The left-hand side is the marginal utility value of extra longevity.
- The right-hand side is the marginal utility cost of using resources for longevity rather than consumption.

Interpretation: This is the ordinary quality-versus-quantity-of-life trade-off. More life is valuable, but it requires resources that could have been consumed.

4.3 Mortality-contingent wealth

With a mortality-contingent claim, wealth depends on realized lifetime, written as $W(T(M))$. The optimization becomes:

$$V(W(T(M)) - M, T(M)).$$

The first-order condition becomes:

$$T_M (V_W W_T + V_T) = V_W.$$

- V_T is the direct utility value of additional life.
- $V_W W_T$ is the additional wealth value created when living longer changes the claim payoff.
- For annuities, $W_T > 0$, so the incentive to extend life is stronger.
- For life insurance paid at death, the sign and the relevant decision-maker can differ.

Interpretation: The new term $V_W W_T$ is the heart of the paper. Mortality-contingent claims alter the marginal return to longevity investment.

4.4 Private annuity pricing and moral hazard

Consider uncertainty states $s = 1, \dots, S$, with probabilities π_s . A person buys an annuity paying Y per year. If longevity in state s is T_s , then the annuity price under competitive pricing and no discounting is:

$$p = \sum_{s=1}^S \pi_s T_s Y = E[T]Y.$$

If the individual chooses longevity investments without internalizing the effect on premia paid by others, private incentives lead to excessive longevity relative to the social optimum.

A social optimum balances the private benefit of longevity in a state with the additional premium cost imposed on the pool. The paper expresses this idea as:

$$\pi_s \frac{dT_s}{dM_s} (V_W Y + V_T - V_W) = \frac{dp}{dM_s} E[V_W].$$

Interpretation: An annuitant who extends life receives more payments, but unless prices fully adjust, part of the cost is borne by the annuity pool. This is moral hazard in longevity investment.

4.5 Public annuities

In a public annuity without behavioral risk rating, longevity-contingent wealth is:

$$W(T) = YT.$$

The marginal wealth impact of an extra year is:

$$W_T = Y.$$

Annual consumption under the annuity can be written:

$$c(t) = \frac{YT(M) - M}{T(M)} = Y - \frac{M}{T(M)}.$$

Interpretation: The annuitized individual does not finance the entire extra year from a fixed wealth stock. A longer life generates more annuity receipts, so longevity is cheaper than under a lump-sum transfer.

4.6 Morbidity-contingent annuity allocation

The paper also considers morbidity states. Let a state s differ by how morbidity affects consumption utility and health production. A state-contingent annuity allocation is $(Y_1, \dots, Y_S)$, and expected utility is:

$$EU = \sum_{s=1}^S \pi_s V(T_s Y_s - M_s, T_s).$$

The budget constraint is:

$$\sum_{s=1}^S \pi_s T_s Y_s \leq W.$$

A key condition can be written:

$$\frac{V_W(s)}{V_W(s')} = \frac{\eta_s + 1}{\eta_{s'} + 1},$$

where $\eta_s = (dT_s/dY)Y/T_s$ is the elasticity of longevity with respect to the annuity level in state s .

Interpretation: Optimal annuity payments should depend on health states if morbidity affects either the value of consumption or the productivity of health investment. A standard annuity that pays the same amount in all morbidity states is generally incomplete.

4.7 PAYG public annuities and private savings

Let L be working years and T retirement years. Let annual after-tax earnings be $W(1 - \tau)$, public annuity benefits be Y , and health investment be $M(Y)$. Smoothed annual consumption is:

$$C = \frac{LW(1 - \tau) + TY - M(Y)}{L + T}.$$

Savings during working years are:

$$S = W(1 - \tau) - C.$$

Using $\rho(Y) = L/(L + T)$, the paper writes savings as:

$$S = [1 - \rho(Y)][W(1 - \tau) - Y] + m(Y),$$

where $m(Y) = M(Y)/(L + T)$.

Interpretation: Higher annuities reduce the need for private saving through a replacement-rate effect, but they can also increase saving by extending life and raising health investment.

4.8 Public annuities and public health care

Let f be the fraction of health care subsidized. Health care demand is $M(f, Y)$, increasing in both the subsidy and the annuity benefit. With a balanced budget for health care and Social Security:

$$NT[M(f, Y)f + Y] = NLW\tau.$$

The implied tax rate is:

$$\tau(f, Y) = \delta(f, Y) \frac{M(f, Y)f + Y}{W}.$$

Interpretation: Medical subsidies and annuity benefits reinforce each other. Health subsidies make longevity cheaper; annuities make longevity more valuable. This creates feedback between the largest old-age public programs.

4.9 Empirical decomposition of annuity value

The price of an annuity of monthly benefit Y is:

$$p(Y) = E[T | Y]Y.$$

With endogenous longevity, the unit price $p(Y)/Y = E[T | Y]$ rises with Y . The change in annuity value can be decomposed as:

$$dp(Y) = p^u(Y)dY + Ydp^u(Y),$$

where $p^u(Y)$ is the unit price. Dividing by total value change gives:

$$1 = \frac{Ydp^u(Y)}{dp(Y)} + \frac{p^u(Y)dY}{dp(Y)}.$$

- The first term is the **incentive component**: the part due to increased longevity.
- The second term is the **quantity component**: the part due simply to a larger annual benefit.

Interpretation: Figure 3 implements this decomposition using survival curves from Figure 2.

5 Identification and empirical design

5.1 What is identified conceptually

- The theoretical model identifies how annuities change the first-order condition for longevity investment.
- The core prediction is qualitative: holding wealth constant, annuitization tilts choices toward longer life relative to a lump-sum payment.
- Public annuities should create larger longevity incentives than fully risk-rated private annuities.
- Health subsidies should amplify the annuity-induced incentive to invest in longevity.

Interpretation: The paper’s strongest contribution is conceptual. It shows that standard old-age support models change once mortality is endogenous.

5.2 Private versus public pricing

- Private insurers may partly internalize longevity moral hazard by risk-rating premia using observable behavior such as smoking or other risk factors.
- Public programs typically do not have the same profit motive or pricing mechanism.
- Mandatory public participation also makes individuals more annuitized than they would otherwise choose privately.

Interpretation: The public-private comparison is central for welfare. A distortion that may be limited by private pricing can be magnified in public annuity systems.

5.3 Evidence and limitations in the CSRS illustration

- The CSRS evidence shows that higher annuity groups have higher survival among males aged 57 and above.
- The data are grouped by salary and annuity cells, which helps but does not fully eliminate confounding.
- Higher annuities may be correlated with lifetime income, education, occupation, or health, all of which could independently affect survival.
- The authors explicitly present the evidence as an illustration of how to measure incentive effects, not as a definitive causal estimate.

Interpretation: The empirical section is best read as a measurement exercise. It motivates future work using richer longitudinal health and retirement data.

5.4 Main empirical obstacle

- Annuity levels are mechanically related to past earnings in many pension systems.
- Past earnings are also related to health, socioeconomic status, and longevity.
- Therefore, separating the substitution effect of annuity form from the income or selection effect of lifetime earnings is difficult.
- The paper suggests that data such as the Health and Retirement Study and Assets and Health Dynamics of the Oldest Old would be useful for future estimation.

Interpretation: The hardest empirical problem is distinguishing whether higher annuity income causes longer life or simply reflects characteristics of people who would live longer anyway.

6 Tables and figures

6.1 Figures 1 and 2: The mechanism and the survival evidence

Figure 1 read:

- The vertical axis is life length T , and the horizontal axis is wealth W .
- The annuitized opportunity set is flatter than the lump-sum opportunity set because longevity generates additional annuity payments.
- The annuitized optimum is labeled (W_A, T_A) , while the lump-sum optimum is (W_N, T_N) .
- Holding overall wealth constant, the annuitized individual chooses greater longevity and lower wealth/consumption quality than under lump-sum wealth.

Figure 1 economic message: Annuities make longevity cheaper. The key distortion is not simply that public pensions make the old richer; it is that the payment form rewards years lived.

Figure 2 read:

- The figure plots survival functions by monthly annuity size for males aged 57–105 in the CSRS data.
- Higher monthly annuity groups have survival curves that lie above lower-annuity groups.
- The figure visually supports the model's premise that higher annuity income is associated with longer survival.
- The authors caution that the relationship may reflect confounding factors, but they use it to illustrate the magnitude of annuity-related feedback effects.

Figure 2 economic message: Even relatively modest survival differences can matter a lot for annuity valuation, because annuity payouts continue for every additional year lived.

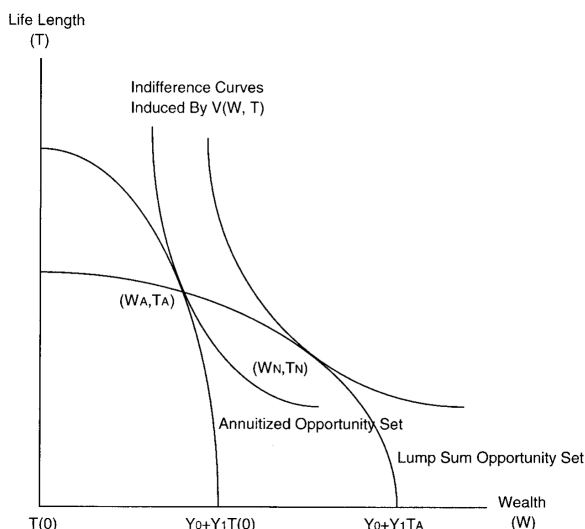


FIG. 1.—Life extension and annuitization

(a) Figure 1: Life extension and annuitization

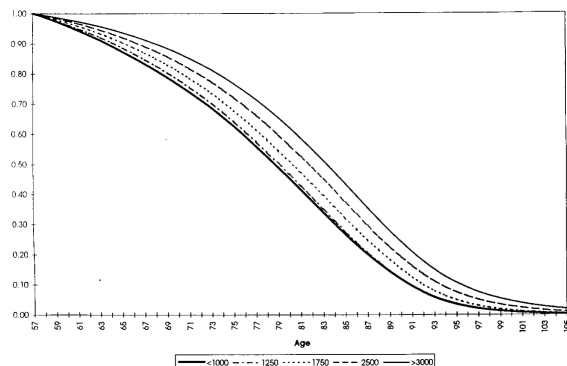


FIG. 2.—Survival functions by monthly annuity size for males aged 57–105

(b) Figure 2: Survival by monthly annuity size

6.2 Figure 3: Incentive effects by monthly annuity increase

Read:

- Figure 3 plots the incentive component of increasing the monthly annuity benefit, by starting age.
- The three curves correspond to larger monthly benefit increases, such as from \$500 to \$1,750, \$2,500, or \$3,500.
- The incentive component is the share of the value increase attributable to the induced increase in longevity rather than the mechanical increase in annual benefit size.
- The top curve is largest because a larger annuity increase produces a larger longevity-linked price effect.
- The paper notes that if an annuity starts at age 70, raising monthly benefits from \$500 to \$2,500 has an incentive component of about 18 percent.

Economic message: The induced-longevity component is quantitatively meaningful. If annuities increase survival, then public pension cost growth partly reflects behavioral feedback, not just mechanically larger benefits.

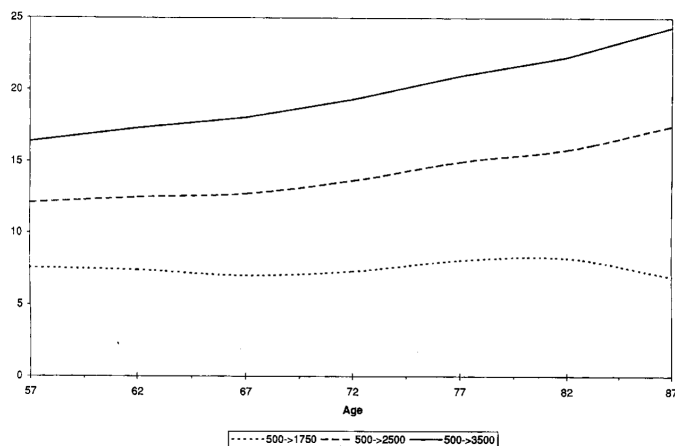


FIG. 3.—Incentive effects by monthly annuity increase for males at age 57–87

Figure 3: Incentive effects by monthly annuity increase

7 Short summary

- The paper studies mortality-contingent claims when longevity is endogenous.
- Annuities reward survival because total benefits increase with length of life.
- This creates moral hazard: annuitants may invest more in longevity than is socially efficient if the cost is pooled or publicly financed.
- Private annuity markets can partly address this problem through risk rating, but public annuities usually cannot.

- Public annuities distort the quality-versus-quantity trade-off toward living longer rather than consuming more per year.
- Standard annuities are incomplete with respect to morbidity; they pay the same annual income in good and bad health states, even though consumption may be more valuable when healthy.
- Public annuities and health insurance interact positively: annuities raise the benefit of longer life, and health insurance lowers its cost.
- With endogenous longevity, public savings programs may raise rather than reduce private savings, because individuals need resources for a longer retirement.
- The CSRS illustration shows higher survival for higher annuity groups and a nontrivial incentive component in annuity valuation.

8 Conclusion

8.1 Core conclusion

Philipson and Becker show that mortality-contingent claims are not behaviorally neutral when longevity is partly endogenous. Annuities insure against outliving assets, but they also create incentives to extend life. This changes the welfare analysis of public pensions, private annuity markets, health subsidies, and savings behavior.

8.2 Economic intuition

The economic intuition is simple. A retiree with a lump sum must finance any extra year of life out of the same wealth stock. A retiree with an annuity receives additional payments when life is extended. Therefore, annuitization makes the marginal cost of longevity lower and shifts choices toward longer life. The same logic explains why health care subsidies and annuities reinforce each other: one raises the return to survival, and the other lowers the price of survival-producing investments.

8.3 Incremental contribution revisited

The paper advances old-age social insurance theory by adding a behavioral margin that standard models suppress. It does not merely say that Social Security affects poverty or consumption smoothing; it says that Social Security changes health investment and mortality behavior. This generates new predictions about public-private savings, public health spending, pension costs, and the distributional consequences of annuity benefits.

8.4 Policy implications

The analysis implies that pension and health policy should be evaluated jointly. A pension reform that changes annuity generosity can affect Medicare-like health spending by changing survival incentives. A health subsidy can affect pension spending by extending the duration of benefit receipt. Policies that look separate in accounting terms may be tightly linked behaviorally.

8.5 Limitations

The empirical section is illustrative rather than definitive. Higher annuity levels may be correlated with lifetime earnings, education, occupation, and other traits that also predict longevity. The model also abstracts from richer health dynamics, heterogeneous preferences over quality of life, family bargaining, and the possibility that multiple people make investments in one person's survival.

8.6 Takeaway

The enduring takeaway is that old-age income programs pay for life, and people can influence the length of life. Once that is recognized, annuities become more than insurance: they are incentives for longevity. This insight changes how we think about Social Security, Medicare, private annuities, savings, and the fiscal consequences of population aging.